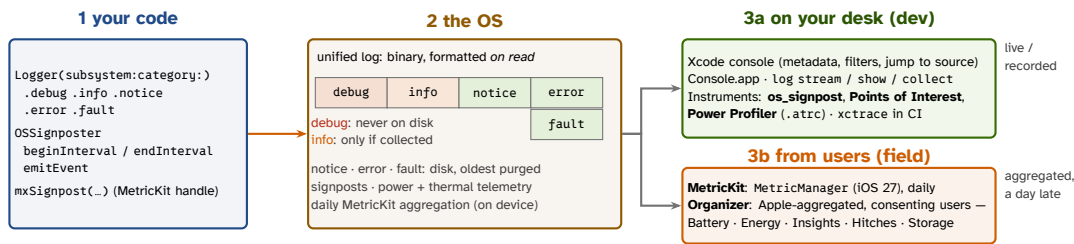


In one line: **Logger** writes *words* into the OS's unified log (binary, levelled, redacted by default), **OSSignposter** writes *time* (intervals and events Instruments draws as tracks), and the OS meters *power* per subsystem. Read it live (Xcode console, Console, Log), recorded (Instruments, Power Profiler, xctrace), or aggregated from users a day later (MetricKit, Organizer).



Unified logging (iOS 10+; Logger iOS 14)

- Logger(subsystem: "com.acme.app", category: "network"): reverse-DNS area + component, both filter keys. Replaced ASL and syslog.
- Stored **binary + compressed**, formatted only when read; a discarded debug is near-free (the compiler skips building its message). `print = stdout`, "for command-line UI": no level, metadata or redaction.

level	persisted (Apple's table)	alias	shown
debug	no — memory only	trace	
info	only if collected with log		
notice	yes, to a storage limit (default)	log	
error	yes + the activity's process chain	warning	yellow
fault	same; the slowest level	critical	red

- Privacy** (.auto): ints, floats, bools shown; dynamic strings and objects **redacted**. `.public` shows; `.private` = `.sensitive` redact; `mask`: `.hash` prints a per-process hash instead, so equal values correlate.
- Xcode 15+ console:** metadata under each line, tokenised filters, Space = Quick Look, jump to the source line. **Console.app:** Action > Include Info / Debug Messages — live stream only; a `.logarchive` already has them.
- Field:** try `OSLogStore(scope: .currentProcessIdentifier)` (iOS 15) reads your own entries back for a report; or a `sysdiagnose`.

Example

```
import os
extension Logger {
    static let net = Logger(subsystem: "com.acme.app",
                           category: "network")
    Logger.net.info("GET \(url.path, privacy: .public)")
    Logger.net.error("HTTP \(code) user \(uid,
                                privacy: .private(mask: .hash))")
    let sp = OSSignposter(logger: .net)
    func fetch() async throws -> Feed {
        let st = sp.beginInterval("Fetch", id: sp.makeSignpostID())
        defer { sp.endInterval("Fetch", st) } // same StaticString name
        sp.emitEvent("CacheMiss")           // a point, not a span
        return try await api.feed()
    }
}

log stream --level debug --predicate 'subsystem == "com.acme.app"'
log collect --device --start 2026-10-07 --output bug.logarchive
xcrun xctrace record --template 'Time Profiler' --time-limit 30s \
    --window 5s --no-prompt --output feed.trace --launch -- MyApp.app
```

Signposts — time, not words

- Interval:** begin/end with the same static name. The returned `OSSignpostIntervalState` enforces one end per begin, same name, same log — a violation **crashes Debug builds**; Release attaches an error to the end.
- ID `.exclusive` (default) if only one such interval runs at a time, else `makeSignpostID()`. **Event** = `emitEvent`, an instant.
- `.pointsOfInterest` category → *Points of Interest*, auto-included with Time Profiler. `OSSignposter.disabled` ≈ no-op; gate costly metadata on `isEnabled`.
- Xcode 27:** a track **per signpost name** under its category; `os_log` overlaid on Time/CPU Profiler thread tracks; `StateReporting` states in *Points of Interest*.
- CI: `XCTOSSignpostMetric(subsystem:category:name:)` in `measure(metrics:)` — no result without a begin/end pair. `xctrace: --run-name (26); --recording-options`, export straight from `.atrc/.logarchive` with `--time-start/--time-end/--duration (27)`.

Power Profiler (Xcode 26; iOS / iPadOS 26)

- Blank template + *Power Profiler*. System track: **% of battery per hour**, charger, thermal state, brightness, SoC sleep. App lanes: **power impact of CPU · GPU · display · networking** — a score; fix the highest first. *All Processes* = system only; app lanes need Launch or Attach.
- No Mac:** Settings > Developer > *Performance Trace* → Power Profiler mode → tick the app → Control Center control; up to **10 h**; share the `.aar` → `.atrc` → Instruments (+ a lower-rate Time Profiler). Testers too.

One subsystem — metric — fix

lane	MetricKit (iOS 27)	fix (Apple's guidance)
CPU	CPUTime, CPUInstructionsCount	events, not polling; lowest QoS that fits; 10–100 ms operations
GPU	GPUPTime	redraw less (<i>Flash Updated Regions</i>); solid backgrounds over blur
display	PixelLuminance (APL 0–100, OLED)	brighter pixels cost more: Dark Mode; stop finished animations
network	TotalWiFi..., TotalCellular... up/down	one URLSession; batch + compress; isDiscretionary (waits for charging or Wi-Fi)
location (no lane)	LocationActivityTime	low desiredAccuracy, distanceFilter; stop when stationary

Field — MetricKit & Organizer

- iOS 27** `MetricManager` (MX... deprecated in 27.2): `logHandle(category:) + mxSignpost` → daily `SignpostIntervalMetric` (count, duration histogram, `cpuTime`, `logicalWrites`, `averageMemory` — the last three only from `mxSignpost`, not `OSSignposter`). Custom signposts are capped — critical paths only. Diagnostics list the signposts active around the event. Test: *Debug > MetricKit > Simulate MetricKit Payloads*.

```
let mk = MetricManager() // iOS 27: keep ONE
let h = MetricManager.logHandle(category: "Feed")
mxSignpost(.begin, log: h, name: "Load") // ... work ...
mxSignpost(.end, log: h, name: "Load")
Task { for await r in mk.metricReports { // daily, never throws
    for case .signpostInterval(let m)
        in r.intervalEntries.fullDayEntry.values { upload(m) } } }
```
- Organizer** (anonymised, consenting users, needs volume). *Battery Usage*: on-screen % per 24 h off power, + background, by Audio · Networking · Processing · Display · Bluetooth · Location · Camera · Torch · NFC. *Energy*: exception reports with a sample stack + 14-day trend.
- Xcode 27:** *Insights Overview*, *Hitches* (all animations, was Scrolling), *Storage*; goals per metric; *Generate Recommendations*.

Traps

- ! Support asks for logs and there is nothing: debug/info weren't persisted — log what support needs at notice+.
- ! privacy: `.public` on a token or email — a PII leak; `.hash` correlates instead.
- ! Power profiled on a charger (cable or MagSafe) reads **0** — pair wirelessly. Impact scores from two device models don't compare.
- ! Thousands of signposts/s recorded *immediately* skip OS buffering — use `--window`.

Remember

Logger = words · signposts = time · Power Profiler = %/h per subsystem · MetricKit / Organizer = the field, a day late.